Nuclear Technology

ISSN: 0029-5450 (Print) 1943-7471 (Online) Journal homepage: www.tandfonline.com/journals/unct20

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To cite this article: Hugh D. Selby, Susan K. Hanson, Daniel Meininger, Warren J. Oldham, William S. Kinman, Jeffrey L. Miller, Sean D. Reilly, Allison M. Wende, Jennifer L. Berger, Jeremy Inglis, Anthony D. Pollington, Christopher R. Waidmann, Roger A. Meade, Kevin L. Buescher, James R. Gattiker, Scott A. Vander Wiel & Peter W. Marcy (2021) A New Yield Assessment for the Trinity Nuclear Test, 75 Years Later, Nuclear Technology, 207:sup1, 321-325, DOI: 10.1080/00295450.2021.1932176

To link to this article: https://doi.org/10.1080/00295450.2021.1932176

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Published online: 16 Nov 2021.

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NUCLEAR TECHNOLOGY · VOLUME 207 · S321–S325 · SUPPLEMENT 1 · 2021
DOI: https://doi.org/10.1080/00295450.2021.1932176

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Technical Note

A New Yield Assessment for the Trinity Nuclear Test, 75 Years Later

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Received February 4, 2021
Accepted for Publication May 14, 2021

Abstract — *New measurement and assessment techniques have been applied to the radiochemical reevaluation of the Trinity Event. Thirteen trinitite samples were dissolved and analyzed using a combination of traditional decay counting methods and the mass spectrometry techniques. The resulting data were assessed using advanced simulation tools to afford a final yield determination of 24.8 ± 2 kt TNT equivalent, substantially higher than the previous U.S. Department of Energy released value of 21 kt.*

Keywords — *Trinity, nuclear tests, radiochemistry.*

Note — *Some figures may be in color only in the electronic version.*

I. INTRODUCTION

This paper is intended to complement the work of Susan Hanson and Warren Oldham, seen elsewhere in this issue.[1] The Trinity nuclear test was fired on July 16, 1945. Its sudden and tremendous power was unlike anything humankind had ever witnessed. Virtually every aspect of the test was developed from whole cloth to study the unknown. The physics, engineering, metallurgy, chemistry, and fabrication that led to the Trinity device and the speed at which they were accomplished were unparalleled at the time and arguably remain so today.

This technical story has captivated scientists and nonscientists alike since that fateful July day. Over the years, no nuclear test has been as thoroughly studied. A simple example of the interest and fascination with Trinity is the fact that trinitite is readily purchased for conducting experiments and pedagogical demonstrations today.[2] Such studies continue to fuel the steady stream of Trinity-based literature 75 years after the test. Yet the yield of Trinity itself has been surprisingly uncertain and has changed several times in the decades between the test and this report.[3–5] These yields range from about 18.6 (±3.7) to 21 (±2) kt TNT equivalent (inside the uncertainty estimate of ~20% 1σ offered by the original Los Alamos Radiochemistry Group). The changes were driven largely by the desire to reexamine Trinity through the lens of more advanced measurement technology, for example, or more detailed evaluation models. Regardless of motivation, all were limited by the developmental nature of the original data set generated in 1945.

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II. RADIOCHEMISTRY

How can this be? How can one of the most lavishly diagnosed nuclear tests, with some of the greatest minds characterizing its performance, not be known to better than about 20%? In brief, the original assessment was limited by sample variability and measurement technology. The challenges these two problems posed is best explained in the context of the Radiochemistry Group's origin.

The Radiochemistry Group was created for the express purpose of determining the efficiency of the Trinity core, or pit, through measurements of residual plutonium fuel and fission fragments. The ratio of these two classes of detonation products is proportional to the efficiency of the plutonium, which is derived from the following expression:

$$\mathrm{Pu}^0 = \mathrm{Pu}^{res} + F \quad , \tag{1}$$

where

Pu^0 = number of ingoing ^{239}Pu atoms in the pit (about 6 kg or 1.15×10^{25} atoms for Trinity; we ignore minor isotopes for reasons discussed later)

$\mathrm{Pu}^{\mathrm{res}}$ = unburned plutonium following the test

F = number of fissions that occurred (alternatively, the number of original plutonium atoms that were converted to fission fragments through the fission process).

Therefore, efficiency can be estimated as the ratio of posttest fissions and ingoing plutonium:

$$E_{\mathrm{Pu}} \sim \frac{F}{\mathrm{Pu}^0} \quad . \tag{2}$$

Substituting the definition of Pu^0 from Eq. (1):

$$E_{\mathrm{Pu}} = \frac{F}{\mathrm{Pu}^{res} + F} \quad . \tag{3}$$

Finally, combining efficiency with the known mass affords plutonium's explosive equivalent yield, as seen in Eq. (4):

$$Y_{\mathrm{Pu}} = M_{\mathrm{Pu}} \times E_{\mathrm{pu}} \times 18.2 \ \mathrm{kt/kg_{pu}} \quad . \tag{4}$$

Equation (4) conveys the concept of burning a known mass of fuel with a given efficiency, much like one might calculate the energy release of a mass of burning hydrocarbon. To determine the conversion constant 18.2 kt/kg in Eq. (4), one first converts M_{Pu} to the number of atoms of ^{239}Pu (Trinity's ingoing isotopic profile contained less than 1% of ^{238,240,241,242}Pu, hence energy release from fission in these isotopes was negligible):

$$N_{^{239}\mathrm{Pu}} = \mathrm{grams}^{239}\mathrm{Pu} \times \frac{1\mathrm{mol}^{239}\mathrm{Pu}}{239.05\mathrm{grams}^{239}\mathrm{Pu}} \times \frac{6.022 \times 10^{23}\mathrm{atoms}}{\mathrm{mol}} \quad . \tag{5}$$

The efficiency E_{Pu} is dimensionless and is expressed in terms of fissions per atom of ^{239}Pu. The total number of fissions is trivially converted to yield in equivalent kilotons of TNT detonation through multiplication by an appropriate energy release/fission value. We used the value of 7.216×10^{-24} kt/f (or 188.4 MeV/f) ^{239}Pu in our recently reported study, for example,[5] taken from Madland, which is in ENDF/B-VII.1 (see Ref. 6, Table XXVII, p. 2954, which is the prompt energy release at fast neutron incident energies for fission fragments, neutrons, and gamma rays).[6]

Equation (3) indicates the type of measurements the Los Alamos Radiochemistry Group made.[7] Peak-yield fission product species such as ^{97}Zr, with high specific activity and high relative production rates (near the peaks of the bimodal fission fragment distribution curve) were excellent proxies for F in the samples. These fragments were strong beta emitters and readily measured in purified samples. Plutonium (Pu^{res}) was quantified with alpha counting methods using aliquots subsampled specifically for the purpose.

A great effort was made to develop the exacting dissolution and subsequent separation chemistries required to make the alpha- and beta-emitter measurements. It is a testament to the Radiochemistry Group that modern radiochemical methods—within and external to the weapons science endeavor—are remarkably similar to those used in 1945. What has changed tremendously is the technology employed to quantify the various radioactivities discussed here. In 1945, the best practical method to quantify beta decay was the use of relatively unstable Geiger-Müller (GM) tubes. These required constant refurbishment and calibration to known quantities of a given beta-emitting species and their corresponding proportionality to fission events.[8] Indeed, one aspect of the nonnuclear "100 Ton Test" that preceded Trinity was to ascertain how well the group could quantify dispersed amounts of known beta activity using GM tubes.[9] Lower activity species (e.g., low-production neutron activation

products or isotopes with long half-lives) were exceedingly difficult to quantify. Many such species were simply not measured in 1945.

Alpha counting was somewhat more robust and relied mainly on Frisch-grid detectors. The low interference rates from alpha emitters in a sample relative to plutonium were a major factor in the comparative ease of plutonium quantification. Still, the sample had to be extremely pure, with minimal residual fission activity remaining. This aspect of plutonium measurement was not really in hand until just after the 100 Ton Test. Taken together, these challenges (and many more) to producing measurements of Pu^{res} and F illustrate the technological contribution to the uncertainty in the original yield estimate of Trinity.

The second factor that contributed to Trinity's uncertainty was the nature of the samples that would be collected. It was well understood at the time that, in order for Eq. (3) to be true, samples had to contain amounts of Pu^{res} and F that represented the actual plutonium efficiency. It was also known that the vapor-phase and condensation chemistries of the various fission fragments would vary significantly relative to plutonium. Consequently, much of the development in the Radiochemistry Group focused on determining which species condensed from the fireball with plutonium and how to collect samples that contained the representative mixture. Two exemplars were the $^{95,97}Zr$ isotopes, which were found to behave in a very similar chemical manner to plutonium. Knowing the chemical compatibility of plutonium and zirconium was only half the battle—it remained to be seen what kind of samples would contain the correct mixture. The sampling question was also part of the purpose of the 100 Ton Test, but no completely satisfactory method was produced before Trinity. Inconveniently, the nuclear detonation had not yet produced samples for method development! Many different techniques were ultimately used to collect samples, including the famous lead-lined tank, rockets, and fallout trays.[9] The various sampling methods unsurprisingly produced varying mixtures of Pu^{res} and F. Careful consideration was given to selecting samples for yield determination without unduly biasing them. The end result was still ~20% variability in efficiency E_{pu} from sample to sample, making sampling the largest contributor to Trinity's uncertainty estimate. As nuclear testing proceeded post war, sampling remained a significant concern for radiochemical yield determination and required years of research to optimize.

Given the difficulties presented by early measurement technologies and suboptimal samples, it is noteworthy how close the original yield estimate (18.6 kt) is to the present accepted value (21 kt). Although our modern assessment (24.8 kt) is significantly higher, the original yield is still within two standard deviations of it—not bad at all, considering the Radiochemistry Group invented the field of weapons radiochemistry in the process. Nevertheless, considerable ongoing debate regarding the true yield of the Trinity test exists in the literature (from 6 to 81 kt in the cited examples), presumably enabled by (1) ready access to trinitite and high-resolution gamma-ray spectrometers and (2) the fairly large uncertainty inferred by the differences in official U.S. Department of Energy (DOE) yield quotations over time.[10,11] The latter is significant, as one would assume that the laboratory that fielded the test would have the most accurate and stable interpretation of performance. Reality is, as just described, complicated by the chemical variability in the original data set. Further, when one considers the fairly short half-lives of the fission fragments that are chemically suitable and diagnostically useful (typically days to a few months), it becomes obvious that there was no way to go back and remeasure them for Trinity, even just a few years after the test. Various Los Alamos National Laboratory (LANL) efforts throughout the years have refined the assessments —hence the updated yield statements—but all of these are ultimately limited by the original data set. Many of the open literature studies emphasize longer half-life fission products like ^{137}Cs ($t_{1/2}$ = 30 year), but these species do not behave chemically like plutonium. This is demonstrated by widely varying activity levels sample to sample and correspondingly large ranges in yield estimates. All of which might suggest that Trinity's yield cannot be more accurately known than it was in 1945.

III. EXTINCT RADIONUCLIDES

Recently, LANL's Nuclear and Radiochemistry Group (the organizational descendant of the original Radiochemistry Group) initiated studies aimed specifically at "resurrecting" the very fission fragments originally used for determining plutonium efficiency in trinitite. This seemingly impossible idea was made possible by the fact that the original fission fragments had not "disappeared." Rather, they simply decayed into stable elements. These stable decay daughters are referred to as extinct radionuclides because they are no longer radioactive and are therefore no longer detectable radiometrically. However, once fixed in a piece of solid trinitite, the short-lived fission fragments quantitatively decay up their beta chains into

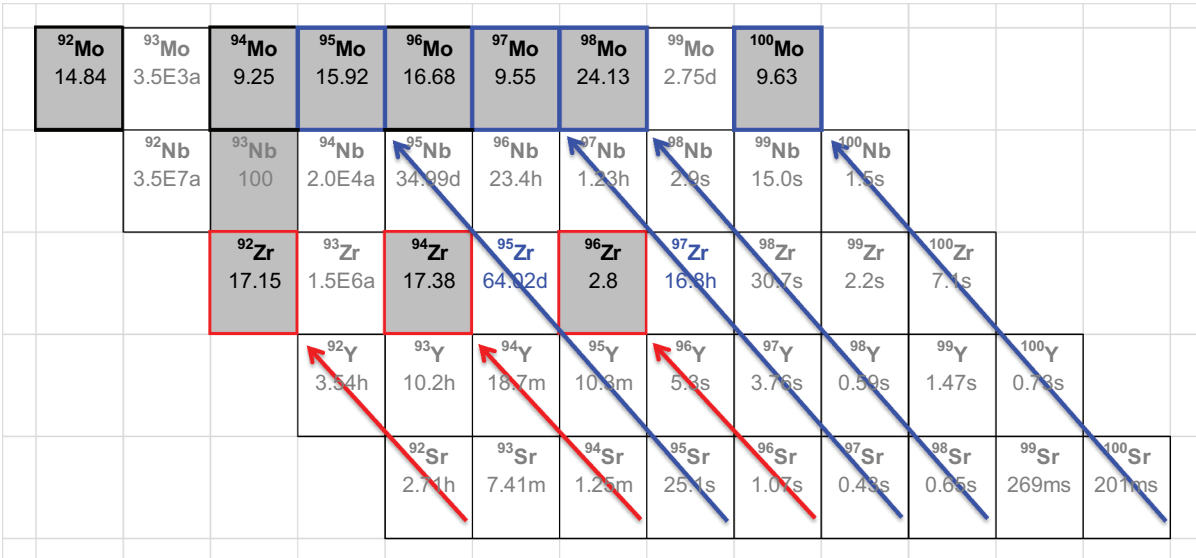

Fig. 1. Partial scheme for the decay of fission products into stable molybdenum and zirconium species. Stable species are denoted with gray boxes, and the natural isotopic abundance is shown for each stable species. Radioactive species and their half-lives are shown in white boxes. Blue and red arrows illustrate β decay chains of interest, and the bolded blue and red boxes indicate stable isotopes that are perturbed by decay of short-lived fission products.

stable isotopes of another element. This subtly perturbs the natural abundances of the chain terminus isotopes in the vitreous solid. A cartoon of this process is shown in Fig. 1. The magnitude of the fission-induced perturbation is minute, typically on the order of a few hundredths of a percent. Until the last decade or so, the precision required to measure this perturbation was not commonly available. Today, high precision mass spectrometric instruments are widespread and LANL successfully employed them to quantify fission-induced $^{95,97}Mo_{Zr}$ in trinitite. The subscript "Zr" refers to the elemental state of the original short-lived radionuclide, with molybdenum being the chain terminus element for the 95- and 97-mass fission fragments. This work was reported by Hanson et al. in the July 2016 edition of the Proceedings of the National Academy of Science.[12]

The Proceedings article was intended to demonstrate the similarity of chemistries between plutonium and zirconium upon condensation, as compared to more readily measured ^{137}Cs. Evidence of plutonium/zirconium co-condensation was made clear by the very consistent ratios of F and Pu^{res}, also measured in the same samples. As a validation of the hypothesis, the plutonium efficiency was calculated using Eq. (3) and the corresponding yield (Eq. 4) was compared to the official DOE Trinity yield. Good agreement was observed (22.7 ± 2.7 kt), and is updated in this report. Although the study was focused on proving a chemical phenomenology hypothesis, it suggested that extinct radionuclides might provide the opportunity for the first complete study of Trinity's yield since 1945.

Unlike the Proceedings study, the new effort was aimed at fully understanding the Trinity device's performance. To this end, 13 samples of trinitite were analyzed for a suite of isotopes, some measured using traditional radiometric techniques, some via the newly developed extinct radionuclide methods. Constructing an accurate picture of Trinity's performance employed what is referred to as the Extinct Radionuclide System (ERS), which folded the measurement data with modern radiochemical and physics simulation tools. The system also included a Bayesian statistical treatment that was developed in parallel with the extinct radionuclide measurements. This ensured a rigorous uncertainty statement for the ERS assessment of Trinity's yield, 24.8 ± 2 kt (Ref. 13).

IV. CONCLUSION

Now, for the first time since the original measurements were made in 1945, we can report a new yield for Trinity derived solely from original trinitite samples using modern experimental and analytical methods. The ERS study shows that Trinity's yield is 24.8 ± 2 kt TNT equivalent, somewhat higher than current and historic estimates. This new assessment's uncertainty overlaps with the previous DOE value of 21 ± 2 kt, where we assigned a conservative 10% relative uncertainty on its value of 21 kt. The source of the accepted DOE yield is not cited in the official DOE publications. However, recently discovered internal communications suggest that the updated yield was the work of Charles

I. Browne, leader of the Radiochemistry Group in 1963 (Ref. 14). The memos are unambiguous regarding the fact that yield assessment was still limited by the original data set. Browne informed his reassessment of Trinity with data from the similar Crossroads tests fired at the Bikini atoll in 1946. While a reasonable approach, the differences in location, height of burst, and myriad other environmental and engineering factors relative to Trinity are all troubling confounds. This suggests Browne's yield is likely less well known than our 10% uncertainty estimate would indicate. In contrast, the present assessment is based on internally consistent data from trinitite samples, all analyzed in the same manner. The data have been evaluated using the most sophisticated modeling tools available, and the resultant models and data subjected to statistical scrutiny. We believe the ERS yield of Trinity is the most accurate and comprehensively characterized since the original Los Alamos Radiochemistry Group's first effort. We suggest that 24.8 ± 2 kt now be adopted as the yield of the Trinity event.

Acknowledgments

The authors gratefully acknowledge the support of the LANL's Laboratory Directed Research and Development investment 20160011DR. We also enjoyed many helpful discussions and edits from M. Chadwick, C. Carmer, K. Spencer, D. Hollis, L. Drake, and D. Mercer. A special thanks is due to B. Archer for his discovery of the memoranda that led to the official DOE yield of Trinity.

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References

1. S. K. HANSON and W. J. OLDHAM, "Weapons Radiochemistry: Trinity and Beyond," *Nucl. Technol.*, **207**, *S1*, S295 (2021); https://doi.org/10.1080/00295450.2021.1951538.

2. D. J. MERCER, P. A. HYPES, and P. R. J. SAEY, "Gamma Ray Measurements of Trinitite and Trinity Site Soil," Weapons Review Letters, **80**, 522 (2021). See also M. P. CROCE et al., "First Measurements of Nuclear Detonation Debris with Decay Energy Spectroscopy," Weapons Review Letters, **80**, 518 (2021).

3. "United States Nuclear Tests July 1945 Through 1981," DOE/NVO-209, Rev. 2, U.S. Department of Energy, Nevada Operations Office, Office of External Affairs (1982).

4. D. W. BARR, "Memorandum to D. Auton," INC-11-603, Defense Nuclear Agency (1983).

5. "United States Nuclear Tests July 1945 Through 1992," DOE/NVO-209 (Rev. 15), U.S. Department of Energy, Nevada Operations Office, Office of External Affairs (2000).

6. D. G. MADLAND, "Total Prompt Energy Release in Neutron-Induced Fission of ^{235}U, ^{238}U, and ^{239}Pu," *Nucl. Phys. A*, **772**, *3–4*, 113 (2006); https://doi.org/10.1016/j.nuclphysa.2006.03.013.

7. J. D. KNIGHT and J. E. SATTIZAHN, "Tracking the Isotopes," Los Alamos National Laboratory (1983).

8. H. D. SELBY et al., "Fission Product Data Measured at Los Alamos for Fission Spectrum and Thermal Neutrons on ^{239}Pu, ^{235}U, ^{238}U," *Nucl. Data Sheets*, **111**, *12*, 2891 (2010); https://doi.org/10.1016/j.nds.2010.11.002. See also M.B. CHADWICK et al., "ENDF/B-VII.1 Nuclear Data for Science and Technology: Cross Sections, Fission Product Yields and Decay Data," *Nucl. Data Sheets*, **112**, No. 12, pp. 2887 (2011).

9. H. L. ANDERSON et al., "Radioactivity Measurements at the 100-Ton Trial," Los Alamos Scientific Laboratory (1945).

10. D. ATKATZ and C. BRAGG, "Determining the Yield of the Trinity Nuclear Device via Gamma-Ray Spectroscopy," *Am. J. Phys.*, **63**, *5*, 411 (1995); https://doi.org/10.1119/1.17902.

11. D. SCHLAUF et al., "Trinitite Redux: Comment on 'Determining the Yield of the Trinity Nuclear Device via Gamma-Ray Spectroscopy, Eds. D. Atkatz and C. Bragg [*Am. J. Phys*. 63 (5),411-413(1995)," *Am. J. Phys*, **65**, *11*, 1110 (1997); https://doi.org/10.1119/1.18793.

12. S. K. HANSON et al., "Measurements of Extinct Fission Products in Nuclear Bomb Debris: Determination of the Yield of the Trinity Nuclear Test 75 Y Later," *Proc. Natl. Acad. Sci.*, **113**, *29*, 29 (2016); https://doi.org/10.1073/pnas.1602792113.

13. H. D. SELBY et al., "Extinct Radionuclides and the Trinity Nuclear Test. Part 2. Debris Diagnostics," accepted for publication in *Def. Res. Rev.* (2021). See also H. D. SELBY, "A Brief History of Radiochemical Diagnostics Seen Through the Lens of the Trinity Nuclear Test," *in press*, Weapons Review Letters (2021).

14. C. I. BROWNE, "Bikini Able and Baker Yields," Internal Memorandum to J. Malik, J-Division, Los Alamos Scientific Laboratory (1963).